

Part D – Recommendations Report for STADIOEquities

1. Introduction

This report's objective is to offer STADIOEquities suggestions derived from the predictive modelling outcomes.. The goal of the modelling is to assist in identifying enrolled clients who might not activate and, more generally, clients who are at risk of becoming inactive within six months. This immediately relates to STADIOEquities since the company has a dormancy issue and an activation gap, and financed activity and client involvement are significant sources of income and assets under management.

Model 1, Logistic Regression, and Model 2, XGBoost, were the two classification models assessed. Accuracy, precision, recall, F1-score, confusion matrices, and ROC-based analysis were used to assess the models.

2. Recommended Model: Model 2 – XGBoost

Model 2 (XGBoost) is suggested as the chosen model for continued development based on actual results.

On the current test set, the two models generated class predictions that were strikingly comparable. The confusion matrix was generated by both:

	Predicted Negative	Predicted Positive
Actual Negative	1 591	1
Actual Positive	2	406

This translates to an accuracy of roughly 99.85%, or 1,997 accurate predictions out of 2,000 observations. Additionally, there are extremely few false positives and false negatives in the results.

However, as the goal of the STADIOEquities project is to identify clients who are at risk of inactivity or non-activation, accuracy alone should not define the final model. As a result, the models' ranking capabilities and probability estimates continue to be

crucial. The ability of XGBoost to represent complex and non-linear relationships and interactions between variables is a benefit. This is especially true for STADIOEquities, where customer behaviour may be influenced by a number of variables, including the frequency of logins, deposits, trading activity, account balance, product usage, and marketing involvement.

Therefore, rather than being a model that should be deployed right away without additional testing, XGBoost is advised as the candidate production model. Because its coefficients offer a relatively simple interpretation of the link between predictors and the expected outcome, logistic regression should be kept as a crucial baseline.

3. Improving the Selected Model

Before XGBoost is taken into consideration for operational use, a number of changes should be made.

The model should first be optimised for hyperparameters. Cross-validation should be used to systematically adjust parameters including learning rate, maximum tree depth, number of estimators, subsampling, and regularisation. This can lessen the chance of overfitting and enhance generality.

Second, a wider range of measures should be used to assess the model. Since a model can attain high accuracy while offering little utility in identifying the minority outcome, accuracy shouldn't be the main metric. Monitoring should be done for accuracy, recall, F1-score, ROC-AUC, and especially precision-recall performance. Additionally, the classification threshold needs to be improved in light of STADIOEquities' operational expenses. For instance, the cost of mistakenly focusing on an active client may be different from the cost of neglecting to detect a customer who later becomes inactive.

Third, explainability and feature importance need to be included. The behavioural and financial traits that most significantly influence individual predictions can be found using methods like SHAP. This would enable STADIOEquities to identify clients who are at risk as well as the observed behaviours linked to that risk.

Lastly, time-based validation should be used to test the model instead of depending solely on a random train-test split. The training period should come before the validation period since the business goal is to prevent future six-month inactivity. This would give a more accurate evaluation of the model's performance in forecasting future customers.

4. Adaptation to the Requested Data

When the real STADIOEquities data requested becomes available, the existing modelling exercise will need to be significantly modified. Compared to a basic static customer dataset, the desired data has far deeper behavioural, financial, and temporal information. Client and account information, app and online behaviour, funding activity, trading activity, product and subscription activity, support interactions, marketing activity, and compliance data are all included in the data request.

The introduction of a temporal prediction framework will be the most significant modification. The model should forecast whether the client will become inactive over the next six months using data collected during a specified observation period. Data leaking could result from including variables that occur beyond the predicted date.

Thus, client-level behavioural traits should be created by combining the event-level data. The number of logins, average session length, quantity and value of deposits, frequency of withdrawals, quantity of trades, frequency of trading, changes in balance, product usage, involvement in campaigns, and recent support interactions are a few examples.

Instead of just capturing static numbers, feature engineering should also record behavioural changes. Reduced trading activity, decreasing deposits, or decreasing login frequency, for instance, could be early indicators of impending dormancy. This strategy is in line with the research, which revealed that behavioural variations and shifts over time were helpful in predicting customer attrition.

Categorical variables including KYC status, acquisition channel, area, platform, transaction type, asset class, and subscription status are also included in the

requested data. These will need to be properly encoded. It is not appropriate to use identifier fields, like CustomerId, as predictive variables.

5. Alignment with Related Literature

The results and the suggested development strategy are generally consistent with the relevant literature that was examined in Part A.

Logistic regression was contrasted with a number of more sophisticated classification algorithms, such as XGBoost and other ensemble techniques, by Hoang, Dang, and Do (2026). Their research places a strong emphasis on feature engineering, class-imbalance treatment, proper preprocessing, preventing data leaking, and model comparison. STADIOEquities can directly benefit from these ideas.

The adoption of a six-month temporal prediction structure is also supported by the literature. Lemos, Silva, and Tabak (2022) forecasted churn over the next six months using past customer data. This is particularly relevant to the goal of STADIOEquities, which is to predict future inactivity.

In a similar vein, Li and Yan (2025) examined class-balancing strategies and model interpretability while comparing a number of ensemble models, including XGBoost. Their methodology backs the suggested application of XGBoost in conjunction with suitable explainability and imbalance control strategies.

Therefore, the STADIOEquities project's direction—comparing various algorithms, building behavioural aspects, limiting leakage, employing suitable assessment measures, and including interpretability rather than just accuracy—is supported by the literature.

6. Conclusion

Logistic Regression should be kept as a baseline and interpretability standard, and XGBoost (Model 2) should be the chosen candidate model based on the results. Although the current findings show good predictive performance on the test data that

is currently available, they should not be taken as proof that production deployment should begin right away.

Incorporating more detailed data, building a true six-month temporal prediction framework, engineering behavioural features, preventing data leakage, addressing class imbalance, fine-tuning the XGBoost model, and testing performance on future-period data should be the main goals of the following phase. In order to translate forecasts into workable client-retention interventions, explainability and threshold optimisation should also be included.

With this strategy, STADIOEquities would be able to move from a proof-of-concept classification model to a more practical decision-support system for identifying customers who would need focused activation or retention treatments.

References

1. Hoang, H.T.T., Dang, T.V. and Do, L.T.D. (2026) 'Customer Churn Prediction in Banking–Machine Learning Application', *Applied Computational Intelligence and Soft Computing*, 2026, Article 8889093.
2. Li, Y. and Yan, K. (2025) 'Prediction of bank credit customers churn based on machine learning and interpretability analysis', *Data Science in Finance and Economics*, 5(1), pp. 19–34.
3. Lemos, R.A.L., Silva, T.C. and Tabak, B.M. (2022) 'Propension to customer churn in a financial institution: a machine learning approach', *Neural Computing and Applications*, 34, pp. 11751–11768.